Project Report

1.) Personal Details

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I am a DS student in this degree currently in diploma level , and this term I took MAD 1 project after completing MAD 1 theory and DBMS last term.

2.) Project Details

Project Statement: Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history.

We were required to build a **Trekking Management Application** web application that allows **Admin, Trek Staff, and Users (Trekkers)** to interact with the system based on their roles.

Approach: I have built a flask python based website to tackle the problem statement given ensuring role based access for users,staff,admin for different access based tasks.

3.) AI/LLM Declaration

I have used chatgpt for :

Testing debugging of my only some amount of python code and logic.

Checking html syntaxes for where it was breaking for eg when it broke for closing div tag and such more smaller work.

Basically I have used chatgpt more for Bootstrap and matplotlib bar chart.

In all I have used around **18-20% of chatgpt in Bootstrap/Jinja/Html** (because I have taken help from my previous project whose bootsrap ui was built by chatgpt) and around **3-5% of chatgpt in failure of some python code**.

4.) Technologies and library Used

Flask :- provides a web development framework for small projects

Flask-login :- user authentication and user management

Flask-sqlalchemy :- orm mapping used to access sqlite db

HTML :- to write frontend ui

Jinja2 :- to connect backend and frontend

Bootstrap :- for ui styling

Sqlite :- light weight database to handle storing different data

Flask-login

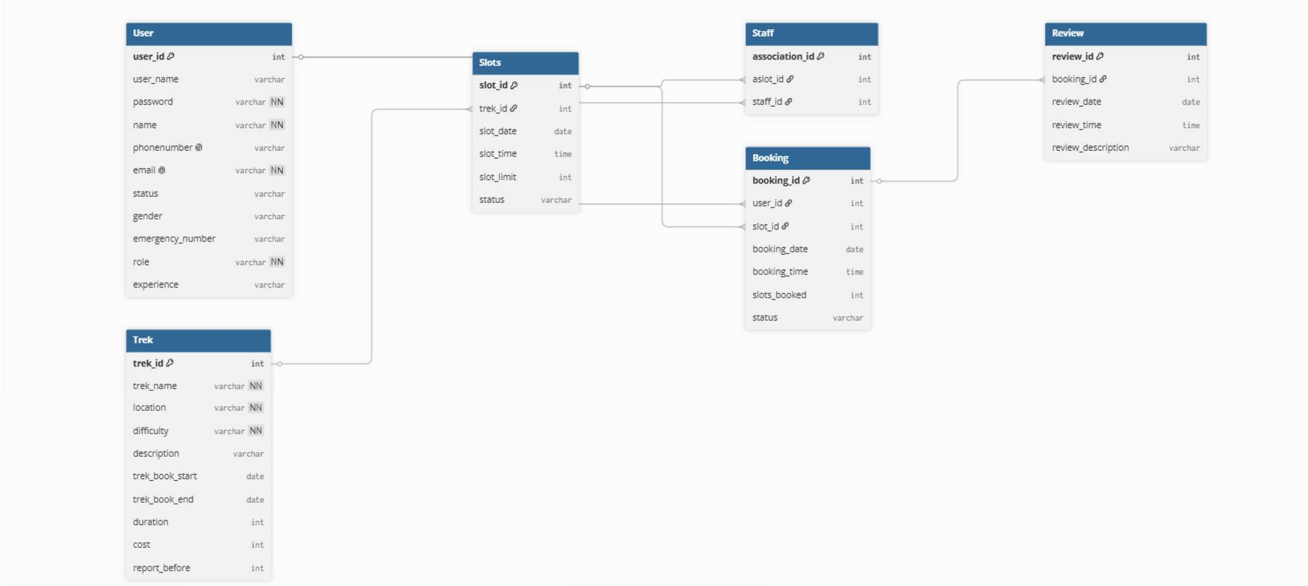
5.) Database Schema / ER diagram

Models:

User: to store info about all users,staff,admin (user_id,user_name,name,password,email,phonenumber,emergency_number,email,role,status,experience)
Trek: to store unique trek info (trek_id,trek_name,location,difficulty,description,trek_book_start,trek_book_end,duration,cost,report_before)
Slots: to store all slots of all treks info (slot_id,trek_id,slot_date,slot_time,slot_limit,status)
Staff: basically association table between staff(saved in user model only) and slots they have been assigned(association_id,aslot_id,staff_id)
Booking: to store all booking details (booking_id,user_id,slot_id,booking_date,booking_time,slots_booked,status)
Review: to store reviews of all bookings (review_id,booking_id,review_date,review_time,review_description)

Relationships:

User-Staff : One to Many
User-Bookings : One to Many
Trek-Slots : One to Many
Slots-Staff :- One to Many
Slots-Booking:-One to Many
Booking-Review:-One to Many
User-slots:-Many To Many(using Staff Model)



6.) Video Representation:-

Link:-
https://drive.google.com/file/d/1kfsZUwqQ22R9WIzupNJeewcfIi8wmuaP/view?usp=sharing

(I may have not been able to show all the implemented features in my video due to 10 min time constraint.)